

#1a: nuclear timescale

$$(i) \quad M_H, \quad f_{nuc} = 0.13$$

$$\rightarrow M_{nuc} = (0.7)(0.13)(M_\odot)$$

$$= 0.091 M_\odot = 0.091 (1.989 \times 10^{30} \text{ kg})$$

$$= 1.81 \times 10^{29} \text{ kg} = 1\% M_\odot$$

$$(ii) \quad \tau_{nuc} = \frac{E_{nuc}}{L} = \frac{\phi M_{nuc} c^2}{L}$$

$$\tau_{nuc} = \phi c^2 (0.091) \frac{M}{L} \left(\frac{1.989 \times 10^{30} \text{ kg}}{M_\odot} \right) \left(\frac{L_\odot}{3.826 \times 10^{26} \text{ W}} \right)$$

$$= \phi c^2 (0.091) \left(\frac{1.989 \times 10^{30} \text{ kg}}{3.826 \times 10^{26} \frac{\text{kg m}^2}{\text{s}^3}} \right) \left(\frac{M}{M_\odot} \right) \left(\frac{L_\odot}{L} \right)$$

$$= (0.007)(2.998 \times 10^8 \frac{\text{m}}{\text{s}})^2 (0.091) (5198.64 \frac{\text{s}^3}{\text{m}^2}) \left(\frac{M}{M_\odot} \right) \left(\frac{L_\odot}{L} \right)$$

$$= 2.98 \times 10^{17} \text{ s} \left(\frac{M}{M_\odot} \right) \left(\frac{L_\odot}{L} \right) \quad \text{yr} = 3.155 \times 10^7$$

$$= 9.4 \times 10^9 \text{ yr} \left(\frac{M}{M_\odot} \right) \left(\frac{L_\odot}{L} \right)$$

$$\approx 10^{10} \text{ yr} \left(\frac{M}{M_\odot} \right) \left(\frac{L_\odot}{L} \right)$$

$$(iii) \tau_{nuc} = 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L_{\odot}}{L} \right)$$

$$\left(\frac{L}{L_{\odot}} \right) = \left(\frac{M}{M_{\odot}} \right)^{3.8}$$

$$\Rightarrow \tau_{nuc} = 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right) \left(\frac{1}{\left(\frac{M}{M_{\odot}} \right)^{3.8}} \right)$$

$$\tau_{nuc} = 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right)^{-2.8}$$

(iv) The nuclear timescale sets the time for a star to exhaust all of its hydrogen fuel.

#1b

$$(i) \tau_{therm} = \tau_{KH} = \frac{GM^2}{2RL}$$

$$\tau_{KH} = \frac{G}{2} \frac{M^2}{RL} \left(\frac{1.989 \times 10^{30} \text{ kg}}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{6.957 \times 10^8 \text{ m}} \right) \left(\frac{L_{\odot}}{3.826 \times 10^{26} \frac{\text{kg m}^2}{\text{s}^3}} \right)$$

$$= \frac{6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg s}^2}}{2} \left(\frac{M^2}{RL} \right) \left(\frac{1.989 \times 10^{30} \text{ kg}}{M_{\odot}} \right) \left(\frac{R_{\odot}}{6.957 \times 10^8 \text{ m}} \right) \left(\frac{L_{\odot}}{3.826 \times 10^{26} \frac{\text{kg m}^2}{\text{s}^3}} \right)$$

$$= 4.96 \times 10^4 \text{ s} \left(\frac{M}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{R} \right) \left(\frac{L_{\odot}}{L} \right)$$

$$= 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{R} \right) \left(\frac{L_{\odot}}{L} \right)$$

$$(ii) \tau_{KH} = 1.5 \text{ e7 yr} \left(\frac{M}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{R} \right) \left(\frac{L_{\odot}}{L} \right)$$

$$\frac{L}{L_{\odot}} = \left(\frac{M}{M_{\odot}} \right)^{3.8} \quad \frac{R}{R_{\odot}} = \left(\frac{M}{M_{\odot}} \right)^{0.7}$$

$$\Rightarrow \tau_{KH} = \frac{1.5 \text{ e7 yr} \left(\frac{M}{M_{\odot}} \right)^2}{\left(\frac{M}{M_{\odot}} \right)^{0.7} \left(\frac{M}{M_{\odot}} \right)^{3.8}} = 1.5 \text{ e7 yr} \left(\frac{M}{M_{\odot}} \right)^{(2-0.7-3.8)}$$

$$\tau_{KH} = 1.5 \text{ e7 yr} \left(\frac{M}{M_{\odot}} \right)^{-2.5}$$

(iii) The thermal timescale sets how long a star takes to regain thermal equilibrium after being disturbed.

#1C

$$(i) \tau_{dyn} = \sqrt{\frac{R^3}{GM}} = (6.67 \text{ e-11} \frac{\text{m}^3}{\text{kg s}^2})^{-1/2} R^{3/2} M^{-1/2}$$

$$\begin{aligned} \tau_{dyn} &= 1.22 \text{ e5} \frac{\text{kg}^{1/2} \text{s}}{\text{m}^{3/2}} \frac{R^{3/2}}{M^{1/2}} \left(\frac{M_{\odot}}{1.989 \text{ e30 kg}} \right)^{1/2} \left(\frac{6.957 \text{ e8 m}}{R_{\odot}} \right)^{3/2} \\ &= 1587 \text{ s} \left(\frac{R}{R_{\odot}} \right)^{3/2} \left(\frac{M_{\odot}}{M} \right)^{1/2} \\ &\approx 0.02 \text{ day} \left(\frac{R}{R_{\odot}} \right)^{3/2} \left(\frac{M_{\odot}}{M} \right)^{1/2} \end{aligned}$$

$$(ii) \tau_{\text{dyn}} \approx 0.02 \text{ day} \left(\frac{R}{R_0} \right)^{3/2} \left(\frac{M_0}{M} \right)^{1/2}$$

$$\left(\frac{R}{R_0} \right) = \left(\frac{M}{M_0} \right)^{0.7}$$

$$\Rightarrow \left(\frac{R}{R_0} \right)^{3/2} = \left[\left(\frac{M}{M_0} \right)^{0.7} \right]^{3/2} = \left(\frac{M}{M_0} \right)^{1.05}$$

$$\rightarrow \tau_{\text{dyn}} \approx 0.02 \text{ day} \left(\frac{M}{M_0} \right)^{1.05} \left(\frac{M_0}{M} \right)^{1/2}$$

$$\tau_{\text{dyn}} \approx 0.02 \text{ day} \left(\frac{M}{M_0} \right)^{0.55}$$

(iii) The dynamical timescale is how fast a star will collapse if the pressure supporting it against gravity were removed.

Another way to say it is:

The dynamical timescale is the time it takes a star to react to being perturbed out of hydrostatic equilibrium

#1d

$$(i) \tau_{\text{nuc}} \simeq 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right)^{-2.8}$$

$$\Rightarrow \tau_{\text{nuc}, 1M_{\odot}} \simeq 10^{10} \text{ yr}$$

$$\tau_{\text{nuc}, 10M_{\odot}} \simeq 10^{10} \text{ yr} (10)^{-2.8} \simeq 1.585 \times 10^7 \text{ yr}$$

$$\tau_{\text{nuc}, 25M_{\odot}} \simeq 10^{10} \text{ yr} (25)^{-2.8} \simeq 1.212 \times 10^6 \text{ yr}$$

$$\tau_{\text{KH}} \simeq 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_{\odot}} \right)^{-2.5}$$

$$\Rightarrow \tau_{\text{KH}, 1M_{\odot}} \simeq 1.5 \times 10^7 \text{ yr}$$

$$\tau_{\text{KH}, 10M_{\odot}} \simeq 4.74 \times 10^4 \text{ yr}$$

$$\tau_{\text{KH}, 25M_{\odot}} \simeq 4800 \text{ yr}$$

$$\tau_{\text{dyn}} \simeq 0.02 \text{ day} \left(\frac{M}{M_{\odot}} \right)^{0.55}$$

$$\Rightarrow \tau_{\text{dyn}, 1M_{\odot}} \simeq 0.02 \text{ day} \simeq 0.5 \text{ hr}$$

$$\tau_{\text{dyn}, 10M_{\odot}} \simeq 0.071 \text{ day} \simeq 1.7 \text{ hr}$$

$$\tau_{\text{dyn}, 25M_{\odot}} \simeq 0.12 \text{ day} \simeq 2.8 \text{ hr}$$

$$(ii) \tau_{dyn} \approx 0.02 \text{ day} \left(\frac{R}{R_0} \right)^{3/2} \left(\frac{M_0}{M} \right)^{1/2}$$

$$\text{if } R = 200 R_0, M \approx 1 M_0 :$$

$$\tau_{dyn, RG} \approx 0.02 \text{ day} (200)^{3/2} (1)^{1/2}$$

$$\boxed{\tau_{dyn, RG} \approx 56.6 \text{ day}}$$

$$\tau_{KH} \approx 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_0} \right)^2 \left(\frac{R_0}{R} \right) \left(\frac{L_0}{L} \right)$$

$$\text{if } R = 200 R_0, M \approx 1 M_0, L = 3000 L_0 :$$

$$\tau_{KH, RG} \approx 1.5 \times 10^7 \text{ yr} (1)^2 \left(\frac{1}{200} \right) \left(\frac{1}{3000} \right)$$

$$\boxed{\tau_{KH, RG} \approx 25 \text{ yr}}$$

$$(iii) \tau_{dyn} \geq \tau_{KH} : \sqrt{\frac{R^3}{GM}} > \frac{GM^2}{2RL}$$

$$\frac{R^3}{GM} > \frac{G^2 M^4}{4 R^2 L^2} \Rightarrow R^5 > \frac{G^3 M^5}{4 L^2}$$

$$\text{check units: } \left[\frac{\text{m}^3}{\text{kg s}^2} \right]^3 [\text{kg}]^5 \left[\frac{\text{s}^3}{\text{kg m}^2} \right]^2$$

$$\left[\frac{\text{m}^9}{\text{kg}^3 \text{s}^6} \right] [\text{kg}^5] \left[\frac{\text{s}^6}{\text{kg}^2 \text{m}^4} \right] = [\text{m}^5]$$

$$R^5 > \frac{G^3 M^5}{4 L^2} ; \text{ but } L = 4\pi\sigma T^4 R^2$$

$$\Rightarrow R^5 > \frac{G^3 M^5}{4} \frac{1}{16\pi^2 \sigma^2 T^8} R^4$$

$$\Rightarrow R^9 > \frac{G^3 M^5}{64\pi^2 \sigma^2 T^8}$$

$$M = 2e30 \text{ kg}$$

$$G = 6.67e-11 \text{ m}^3/\text{s}^2\text{kg}$$

$$\sigma = 5.67e-8 \frac{\text{W}}{\text{m}^2\text{K}^4}$$

$$T = 5770 \text{ K}$$

$$R > \left[\frac{G^3 M^5}{64\pi^2 \sigma^2 T^8} \right]^{1/9} = 2.5 \times 10^{11} \text{ m}$$

$$= 360 R_{\odot}$$

$$\rightarrow \boxed{R > 360 R_{\odot}}$$

$$\tau_{\text{dyn}} = \sqrt{\frac{R^3}{GM}} = 1e7 \text{ s} \sim \frac{1}{3} \text{ yr}$$

$$\tau_{\text{clt}} = \frac{GM^2}{2RL} = 1e7 \text{ s} \sim \frac{1}{3}$$